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Surgical Outcomes of anomalous aortic origin of coronary arteries: early and mid-term results from a single-center experience

Yuefeng Cao^{1†}, Jun Dong^{1†}, Gang Li¹ and Qiang Wang^{1*}

Abstract

Background Anomalous aortic origin of a coronary artery (AAOCA) is the second leading cause of sudden cardiac death in young athletes. While anomalous aortic origin of the right coronary artery (AAORCA) predominates, anomalous aortic origin of the left coronary artery (AAOLCA) carries higher mortality. Surgical correction requires anatomical precision due to heterogeneous anatomy.

Methods A retrospective analysis (June 2016–June 2023) included 17 AAOCA patients (mean age: 11.9 ± 3.8 years). Surgical techniques included unroofing (94.1%), coronary reimplantation (11.8%), pulmonary translocation (5.9%). Preoperative computed tomography angiography (CTA) assessed take-off angles and anatomy. Outcomes included mortality, reoperations, and echocardiographic parameters.

Results Symptomatology included exertional chest pain (35.3%) and syncope (52.9%). AAORCA predominated (64.7%), with 52.9% of coronaries originating above the sinotubular junction. Unroofing significantly increased coronary take-off angles (pre-op: $12.7^\circ \pm 5.9^\circ$ vs. post-op: $38.7^\circ \pm 11.6^\circ$, $p < 0.05$). There was no in-hospital mortality. At the mean follow-up (50.5 ± 21.6 months), survival was 100% with preserved LVEF ($67.6\% \pm 4.1\%$) and no aortic regurgitation. Two reinterventions occurred: one stent implantation for left main coronary artery stenosis and one mitral valve repair. One patient experienced recurrent angina.

Conclusions Unroofing, reimplantation, pulmonary translocation yield excellent early/mid-term survival and preserved ventricular function in AAOCA. Correction of take-off angle is critical for preventing residual ischemia. Individualized surgical planning based on anatomical variants optimizes outcomes.

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Introduction

Anomalous aortic origin of a coronary artery (AAOCA), first documented during the Renaissance period, represents the second most common cause of sudden cardiac death in young athletes. With an estimated prevalence of 0.2% – 1.33% in the general population, AAOCA manifests predominantly as anomalous aortic origin of the right coronary artery (AAORCA), which occurs 3- to 9-fold more frequently than anomalous aortic origin of the left coronary artery (AAOLCA) [1, 2]. Notably, despite its lower incidence, AAOLCA constitutes the predominant etiology of sudden cardiac mortality.

AAOCA demonstrates significant anatomical heterogeneity, with the anomalous coronary artery potentially originating from the ascending aorta, contralateral sinus of Valsalva, or non-coronary sinus. Unroofing represents the most employed surgical technique. Owing to this substantial anatomical variability, alternative approaches such as pulmonary artery translocation are increasingly utilized and demonstrate generally favorable outcomes. Nevertheless, a subset of patients may experience recurrent angina postoperatively. Comprehensive preoperative delineation of AAOCA anatomical features is therefore essential for tailoring optimal surgical management.

The purpose of this study was to present our surgical experience with early and mid-term results in patients operated for AAOCA.

Patients and methods

Patients

This retrospective study included patients who underwent surgery for AAOCA at a single center from June 2016 to June 2023. Medical records included patient demographics, preoperative data, anatomical data, surgical management data, and postoperative outcomes. In-hospital mortality was defined as death occurring within 30 days of surgery or before hospital discharge. The follow-up data included mortality, complications, and echocardiographic parameters. The echocardiographic parameters included left ventricular ejection fraction (LVEF) and coronary ostium morphology. The CTA was performed to assess the location and height of the coronary artery opening, take-off angle. All procedures performed in this study were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards. This study was approved by the institutional research committee of Beijing Anzhen Hospital. We are permitted to collect data for audit and research purposes.

Surgical technique

All operations were performed via median sternotomy using cardiopulmonary bypass (CPB). Cardiac arrest

after aortic cross-clamping (ACC) was achieved with antegrade Custodiol cardioplegia solution. Surgery was carried out using conventional CPB at 28–32 °C.

Unroofing procedure was the most frequent technique (16/17, 94.1%), followed by coronary reimplantation (2/17, 11.8%), pulmonary translocation (1/17, 5.9%). The unroofing procedure was performed using partial transverse aortotomy above the sinotubular junction, the location of coronary ostia and orientation of coronary inspected by coronary probe. After determining the length of the intramural segment, a portion of the inner aortic wall at the coronary artery ostium was cut using 15-degree forward-facing scissors. A neo-ostium was then created by securing the coronary artery ostium to the aortic wall with interrupted 7–0 polypropylene sutures. During the surgical procedure, it is essential to first explore the relationship between the intramural course and the intercoronary commissure to avoid causing aortic valve insufficiency. Additionally, the operation should strive to minimize aggressive unroofing to prevent the incision from penetrating the aorta, which could result in troublesome bleeding. Moreover, aortic commissural resuspension was not performed in any patient during the unroofing surgery.

Coronary reimplantation was utilized in patients lacking an intramural course or with a very short intramural segment, where unroofing was insufficient to relocate the ostium from the interarterial course to the correct sinus of Valsalva (Fig. 1).

Pulmonary translocation was performed in one patient with AAOLCA, in whom the left coronary artery originated from the right sinus of Valsalva and followed a purely interarterial course without an intramural segment. Initial coronary reimplantation was attempted but resulted in significant coronary compression preventing separation from CPB, necessitating concomitant pulmonary translocation.

The choice of surgical technique was individualized based on preoperative CTA findings and intraoperative inspection. Unroofing was performed when an intramural segment was identified and was deemed the preferred approach due to its safety and effectiveness. Coronary reimplantation was indicated in patients with a very short or absent intramural segment, where unroofing would not adequately relocate the ostium to the correct sinus. Pulmonary translocation was utilized in select cases with a purely interarterial course and no intramural segment, particularly when other techniques were not feasible or resulted in coronary compression.

Statistical analysis

All the data were analyzed by using SPSS 24.0 software. Quantitative data are presented as the mean, standard deviation or median (range). Qualitative values are

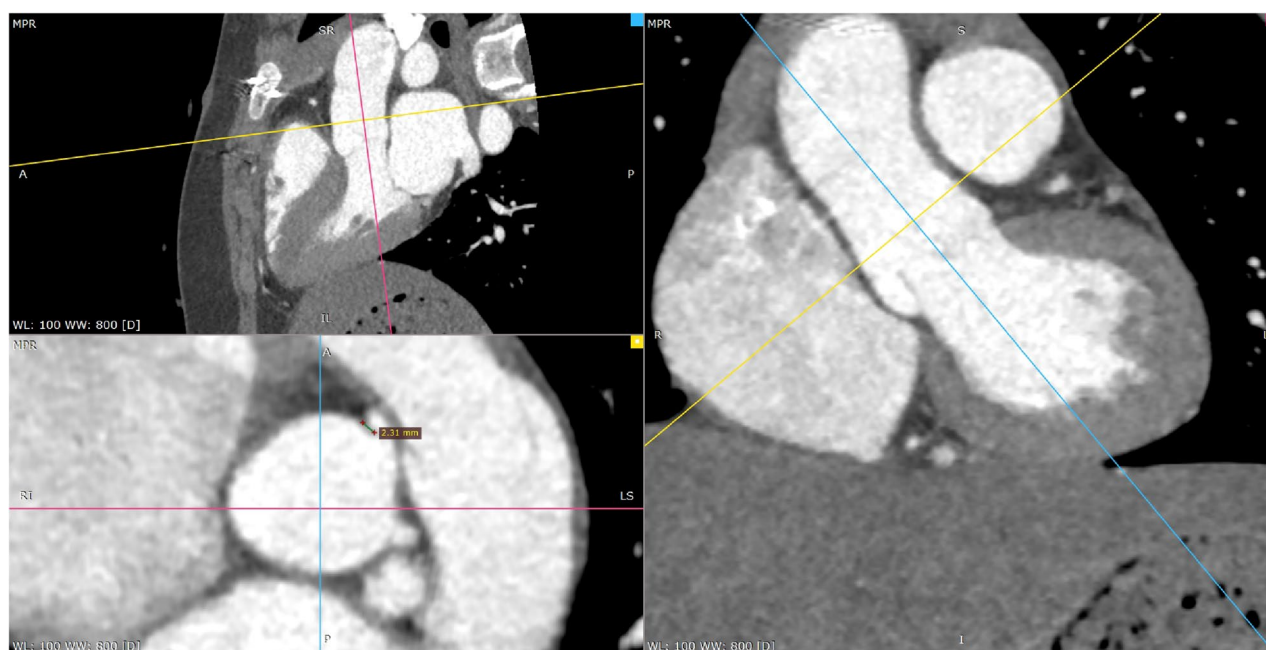


Fig. 1 Anomalous aortic origin of the right coronary artery with short intramural segment

expressed as percentages. All tests were performed in parallel, and $p < 0.05$ was considered to indicate statistical significance.

Results

Perioperative data

During the past 8 years in our center, 17 patients (10 males and 7 females) underwent AAOCA repair. The baseline characteristics are described in Table 1. Six patients (10–14 years) presented with symptoms of exertional chest pain. Nine patients (4–16 years) had also experienced exertional syncope. No chromosomal abnormalities or cardiac malformations such as tetralogy of Fallot or hypoplastic left heart syndrome were observed in this patient cohort.

In this cohort, an intramural course was identified in 16 patients, while 1 patient had an extramural course without an intramural segment. Among the 16 patients with an intramural course, the intramural segment was located above or at the level of the aortic valve commissure in 12 patients (75%) and below the commissure in 4 patients (25%). In all cases, the surgical repair preserved commissural integrity, and no commissural detachment or resuspension was required.

Based on the Angelini et al. classification [3], the cohort included 6 cases of AAOLCA and 11 cases of AAORCA. In this group, one case exhibited a single coronary ostium (Fig. 2), another case had two separate but adjacent ostia, and the remaining patients all had individual orifices. In cases of anomalous coronary artery origins, 9 (52.9%) were located above the sinotubular junction. In this study

cohort, intramural course was identified in 16 patients. The mean take-off angle of the proximal anomalous coronary artery before surgery was $12.7 \pm 5.9^\circ$, and the mean take-off angle after surgery was $38.7^\circ \pm 11.6$ (Fig. 3). In this study, two cases were associated with other cardiac anomalies, including one case with supravalvular aortic stenosis and one case with moderate mitral regurgitation. The patient with mitral regurgitation presented with pre-operative mitral annular dilation, while maintaining normal leaflet morphology, subvalvular apparatus integrity, and left ventricular function. Mitral annular plication was performed as a concomitant surgical intervention.

The median age at the time of repair was 11.9 ± 3.8 years. The median CPB-time was 90 (IQR: 70–98.5) min, and median ACC-time was 46 (IQR: 40.5–64.5) min, median length of mechanical ventilation time was 4 (IQR: 4–6) hours. In this group of patients, 16 underwent unroofing procedure, and 2 underwent coronary reimplantation. No in-hospital mortality was observed in this study.

A 13-year-old boy received extracorporeal membrane oxygenation (ECMO) treatment preoperatively due to myocardial infarction and underwent unroofing procedure after stabilization. This patient demonstrated pre-operative and postoperative take-off angles of 10.3° and 45.7° , respectively. He was readmitted at 1 month and 6 months post-discharge due to recurrent chest pain. Another 15-year-old female patient with AAOLCA was admitted to the hospital due to cardiac arrest. During the surgery, it was found that her intramural segment was very short. She underwent coronary reimplantation with

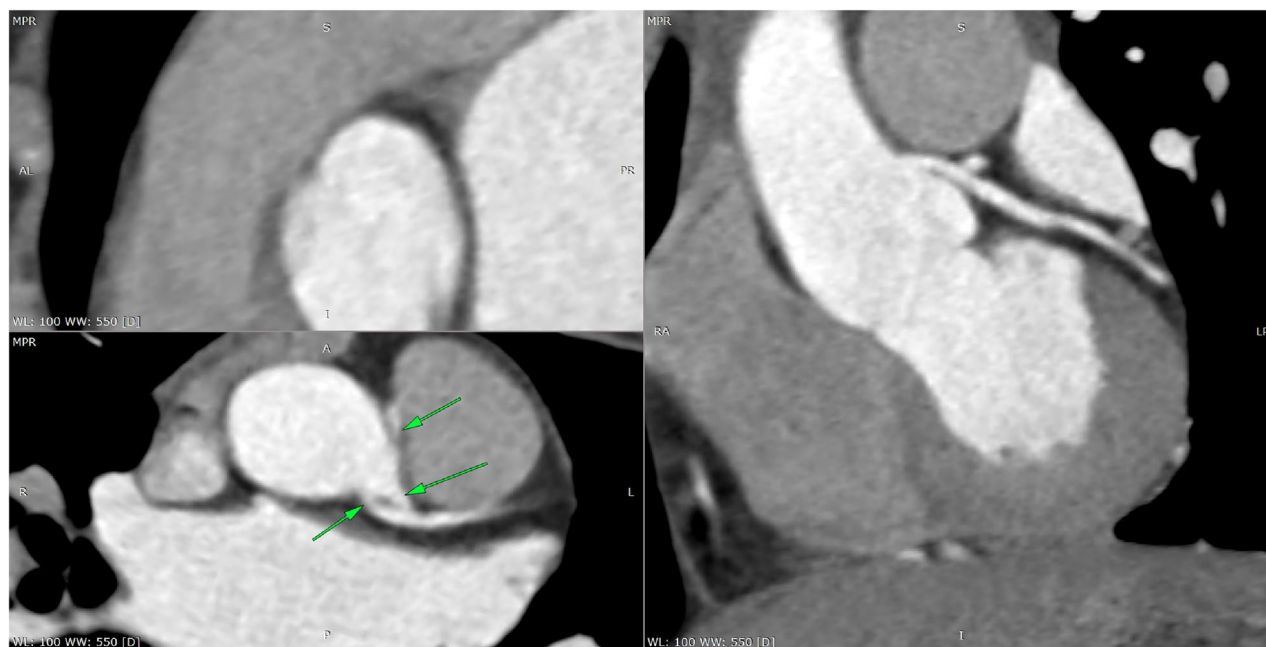
Table 1 Patients' characteristics and outcomes between infant and non-infant groups ($n = 17$)

Patient	Age (years)	symptom	Type	Ostial Relationship	Ostial height	surgery	CPB-time
1	10y11m	chest pain	L	Grade 1	II	unroofing	69
2	13y11m	syncope	L	Grade 1	II	unroofing	73
3	15y11m	cardiac arrest	L	Grade 1	I	*	358
4	15y	Syncope	L	Grade 1	II	unroofing	66
5	13y	Syncope	L	Grade 1	IV	*	146
6	11y	Syncope	L	Grade 1	III	unroofing	90
7	14y	syncope	L	Grade 1	II	unroofing	77
8	14y6m	syncope	R	Grade 1	III	unroofing	57
9	14y9m	chest pain	R	Grade 1	III	unroofing	117
10	14y10m	myocardial infarction	R	Grade 1	II	unroofing	70
11	6y	Syncope	R	Grade 3	IV	*	99
12	16y	Syncope	R	Grade 1	III	unroofing	94
13	5y	chest pain	R	Grade 1	IV	unroofing	80
14	10y	chest pain	R	Grade 1	III	unroofing	94
15	10y	chest pain	R	Grade 1	II	unroofing	93
16	15y	chest pain	R	Grade 2	II	unroofing	98
17	4y	Syncope	R	Grade 1	IV	unroofing	70

Grade 1: two separate ostia; grade 2: separate but adjacent ostia; grade 3: common ostium with bifurcation within the aortic wall; grade 4: single coronary with bifurcation outside of the aortic wall

Ostial height: I centrally located; II high but below the sinotubular junction; III at the sinotubular junction; IV above the sinotubular junction

Concomitant operation: 3*: coronary reimplantation + pulmonary translocation; 5*: unroofing + Mitral Valve Repair; 11*: unroofing + Aortic Valve Repair + Supravalvular Aortic Repair

**Fig. 2** A single coronary ostium

pulmonary translocation procedure, and developed chest pain one month postoperatively. Emergency coronary angiography revealed left main coronary artery stenosis. Given concerns regarding competitive flow with coronary artery bypass grafting (CABG) and the risks associated with reoperation, stent implantation was performed.

Follow-up

The mean follow-up duration was 50.5 ± 21.6 months. There was no late mortality in this study. Follow-up echocardiography demonstrated the LVEF of $67.6\% \pm 4.1\%$. No patients developed aortic regurgitation during follow-up. A patient with AAORCA and a short intramural segment treated only with unroofing visited the

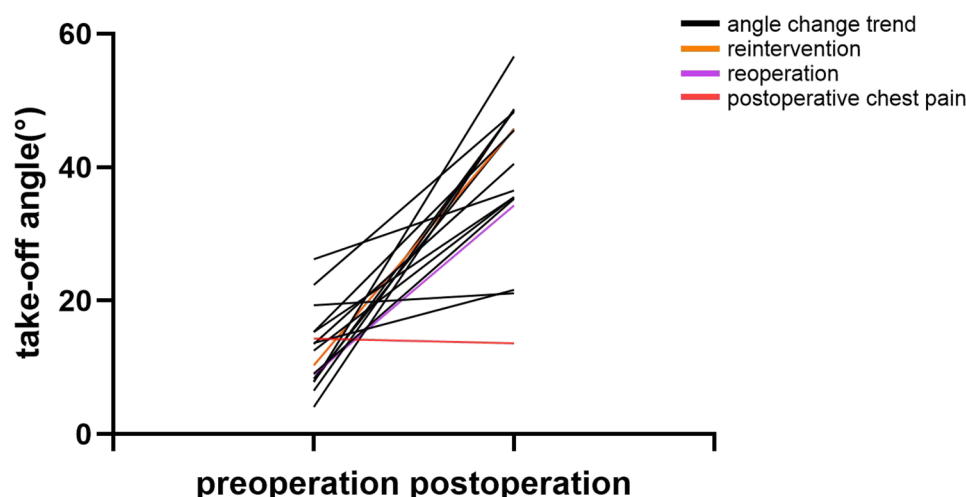


Fig. 3 The take-off angle of the proximal anomalous coronary artery before and after the operation. The two patients requiring reintervention both demonstrated adequate postoperative take-off angle correction. Their clinical courses highlight that factors beyond the take-off angle, such as pre-existing extensive myocardial injury or postoperative focal coronary stenosis, can dictate the need for further intervention. This underscores the multifactorial nature of postoperative outcomes in AAOCA repair. The patient with postoperative chest pain demonstrated inadequate correction of the take-off angle

emergency department twice for chest pain postoperatively. Although CTA indicated inadequate correction of the take-off angle, electrocardiograph was stable and troponin levels were normal, and the patient remains under regular follow-up. During the follow-up period, two patients required further intervention. The first was a percutaneous reintervention: a patient with AAOLCA (status post coronary reimplantation and pulmonary translocation) underwent left main coronary artery stenting for stenosis. The second was a surgical reoperation: another patient with AAOLCA underwent mitral valve repair for four years postoperatively due to significant mitral regurgitation.

Discussion

AAOCA is a rare congenital cardiac anomaly with high importance in clinical practice due to the association with myocardial ischemia and sudden cardiac arrest/death in young patients. AAOCA is commonly found in patients with ischemic symptoms. However, with recent advances in imaging technology, an increasing number of asymptomatic cases have been identified and treated, thereby reducing cardiovascular events. Surgical interventions such as unroofing have become the primary treatment modality with favorable outcomes. In contrast to previous literature, no evidence of postoperative aortic valve regurgitation was observed in this cohort. Nevertheless, some patients still experienced persistent chest pain, which may correlate with their specific anatomical variations, especially the take-off angle. Comprehensive preoperative evaluation can help minimize postoperative complications.

All 17 patients in this study presented with varying degrees of symptoms at initial consultation. Surgical

intervention was recommended for all AAOLCA cases and actively advocated for AAORCA patients with clinical symptoms, which aligns with the treatment philosophy of Mayo Clinic [4]. This study involved multiple surgeons, and the surgical approach was primarily determined by the operating surgeon. However, unroofing surgery was the preferred technique as it proved to be safe and effective for the majority of patients. Although CABG has been widely adopted in the surgical management of AAOCA, concerns persist regarding competitive flow from the internal mammary artery. Moreover, given that all patients in this cohort were minors without concomitant coronary atherosclerotic plaques, CABG was not employed as a surgical approach in our institutional practice.

The unroofing procedure, first reported by Mustafa and colleagues in 1981, continues to be the most common surgical approach for managing the intramural coronary segment in AAOCA patients. It can be performed safely with excellent results in the majority of patients [4]. This technique offers several advantages: it relocates the functional orifice to the appropriate sinus, significantly enlarges the ostium, eliminates the intramural arterial component, and eliminates the risk of ischemia related to the interarterial course. But it also has some particular pitfalls postoperatively, such as the risk of residual stenosis and aortic valve regurgitation [5]. Some surgeons prefer routine autologous pericardial patch augmentation at the coronary ostium to mitigate the risk of residual stenosis in the distal intramural segment [6]. In our study, only one patient who had AAOLCA with a separate ostium and short intramural course underwent an unroofing procedure with patch augmentation, and then received left coronary artery stent implantation after one month

due to left coronary artery stenosis and was prescribed lifelong aspirin therapy. No instances of restenosis were observed in the cohort during follow-up time. However, autologous pericardial patches may actually increase the risk of restenosis attributable to progressive calcification.

Aortic valve functional preservation constitutes a major surgical consideration, as progressive aortic regurgitation remains an important indication for continued postoperative morbidity and need for reintervention [7, 8]. Because the aortic valve commissures are always in close proximity to the intramural coronary course, some surgeons advocate combined unroofing with commissural suspension to optimize valvular function. Yerebakan's study suggested that at a median follow-up of 1.02 years, the unroofing commissural group demonstrated preserved aortic valve function. But our study demonstrated that none of the patients in this series developed aortic regurgitation during the postoperative follow-up period of 50.5 ± 21.6 months, even though commissural suspension was not performed as an adjunct to unroofing procedures. This is similar to the results of another study [9].

In this study, one patient underwent coronary reimplantation with pulmonary translocation procedure. The patient had been hospitalized due to cardiac arrest. After performing an unroofing procedure, weaning from CPB was unsuccessful, so he underwent pulmonary translocation surgery and was successfully weaned off CPB. Unroofing procedure can increase the size of the ostium and essentially moves it to the correct sinus. However, some specific techniques were employed for certain types of AAOCA, such as pulmonary translocation. If the intramural segment is relatively short (< 5 mm), the procedure may fail to relocate the ostium to the correct sinus and/or the anomalous vessel may be at risk of compression by the intercoronary pillar [10]. The pulmonary translocation technique is mainly used when the anomalous coronary is not intramural but the ostia are in close approximation or if there is a single coronary artery in the presence of ischemia [5]. But as demonstrated in the case from this study, this surgical technique could also serve as an adjunctive approach in certain special circumstances.

Abnormal coronary take-off angles, particularly acute take-off angles, are among the key anatomical factors contributing to myocardial ischemia and sudden cardiac death. These anomalies predispose the coronary artery to compression during exercise or hemodynamic changes, leading to myocardial ischemia or even sudden death [11, 12]. CTA serves as the primary modality for evaluating abnormal coronary take-off angles [13]. Some investigators have reported subclinical ischemic manifestations in a subset of patients after unroofing procedures alone. This is attributed to residual compression risks when the coronary artery remains in the incorrect sinus

post-unroofing, especially in cases with a short intramural segment [11, 14]. Correcting the take-off angle is a critical component of anatomical repair strategies, as it helps restore a near-normal coronary anatomy, thereby offering superior long-term protective outcomes [12]. In our study, one patient with a short intramural segment who underwent only unroofing had persistent postoperative ischemic symptoms, presumably due to failure in achieving an adequate anatomical correction of the take-off angle after other causes of ischemia were ruled out [15–18]. Improvement in the take-off angle was associated with postoperative symptom relief. For patients with suboptimal angular correction and persistent symptoms, close clinical follow-up is essential, with reintervention considered when necessary. Evaluating the improvement of postoperative ischemic symptoms remains a clinical challenge, though imaging examinations and stress testing can provide valuable assistance. While our findings support the importance of correcting the take-off angle, we acknowledge the limitation of not establishing a definitive optimal cutoff value. In the literature, a normal coronary take-off angle is generally considered to be greater than 45° , and angles below this threshold are classified as acute take-off [19–21]. Some studies have further suggested that angles $< 30^\circ$ may carry an even higher risk of sudden cardiac death [20, 22]. However, due to the small sample size ($n = 17$) and limited number of adverse events in our cohort, a receiver operating characteristic curve analysis to determine an evidence-based threshold associated with freedom from ischemia was not statistically feasible. Larger, multicenter prospective studies are warranted to validate any potential cutoff value and to establish more precise imaging criteria for surgical success.

Limitations

This study reflects a single-center experience with a relatively small cohort over an extended period. The involvement of multiple surgeons, while reflecting real-world practice, may introduce variability. Centralizing these complex cases to a limited number of high-volume surgeons could potentially optimize consistency and outcomes, a consideration for future program development.

Conclusions

Surgical repair of AAOCA demonstrates favorable early and mid-term outcomes, with 100% survival and preserved left ventricular function at a mean follow-up of 50.5 months. The unroofing technique effectively corrects coronary take-off angles, reducing ischemic risk. However, anatomical complexity—such as short intramural segments or non-intramural interarterial courses—may necessitate adjunctive strategies to prevent residual stenosis. Postoperative angina in select cases highlights

the importance of comprehensive preoperative CTA assessment to optimize anatomical correction. Aortic valve function remained intact without commissural suspension, challenging routine use of this adjunct. Future studies should focus on long-term coronary patency and ischemia prevention in high-risk subtypes.

Abbreviations

AAOCA	Anomalous aortic origin of a coronary artery
AAORCA	Anomalous aortic origin of the right coronary artery
AAOLCA	Anomalous aortic origin of the left coronary artery
CTA	Computed tomography angiography
LVEF	Left ventricular ejection fraction
CPB	Cardiopulmonary bypass
ACC	Aortic cross-clamping
ECMO	Extracorporeal membrane oxygenation
CABG	Coronary artery bypass grafting

Authors' contributions

Yufeng Cao and Jun Dong contributed equally to study design, data analysis, manuscript drafting, and critical revisions. Gang Li contributed to clinical data acquisition and interpretation. Qiang Wang supervised the research, provided funding, and approved the final manuscript. All authors reviewed and endorsed the findings.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

This study was approved by the Ethics Committee of Beijing Anzhen Hospital (Approval No. 2025-0521) and was conducted in accordance with the Declaration of Helsinki. Written informed consent was obtained from all participants or, for participants under 16 years of age, from their parents or legal guardians prior to inclusion in the study.

Consent for publication

Written informed consent for publication of the study results, including any individual person's data, was obtained from all participants or, for participants under 16 years of age, from their parents or legal guardians.

Competing interests

The authors declare no competing interests.

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